

CO5 - AT1 - LEVEL - 3

Level Exam </> C 03 Sep 2026, 10:00 AM

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Result Available
Your score: 63 / 100

QUESTIONS

LEVEL 3 — HARD

Q1
function

Q2
functionsandpointers

2 / 2 saved

Level 3 — Hard matrix (H) — function

Write a function `int sumOfDiagonals(int matrix[][10], int n)` that calculates the sum of the diagonal elements of a given square matrix of size `n x n`.

Sample Input:

Enter the size of the matrix: 3

Enter elements of the matrix:

1 2 3

4 5 6

7 8 9

Sample Output:

Sum of diagonal elements is 15

Your Code

```
1 #include <stdio.h>
2 int sumOfDiagonals(int matrix[][10], int n) {
3     int sum = 0;
4     for (int i = 0; i < n; i++) {
5         sum += matrix[i][i];
6     }
7 }
```

Program Input (stdin)

Enter input if required...

Output

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C PROGRAMMING

M Hemanth reddy

192312534RC

CO5 - AT2 - Debugging

Level Exam

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QUESTIONS

LEVEL 1 — EASY

Q1

Number

Q2

Number

Q3

Number

Q4

Number

4 / 4 saved

Level 1 — Easy

Q19 — Number

```
<PRE>#include<stdio.h>int main(){float x = 5/2;printf("%f\n", x);return 0;}</PRE>
```

Your Code

```
1 #include<stdio.h>2 int main() {3     float x = 5.0/2;4     printf("%f\n", x);5     return 0;6 }
```

Program Input (stdin)

Enter input if required...

Output

2.500000

Run

Save

35°C

Mostly sunny

Search

ENG IN

13:38

03-09-2026

CO5 - AT3 - HACKATHON

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QUESTIONS

Q1

CO5 - AT3 - HACKATHON /
MINIPROJECT
100 marks

1/1 answered

Q1 CO5 - AT3 - HACKATHON / MINIPROJECT

100 marks

CO5 – AT2_HACKATHON

1. A satellite continuously sends telemetry packets containing temperature, battery level, fuel status, altitude, and communication status to a ground station. Thousands of packets may arrive simultaneously from different processing channels. The system must store incoming records dynamically because the number of packets is unknown in advance.

Develop a C-based Real-Time Satellite Telemetry Processing System that:

- Uses dynamic memory allocation for telemetry records.
- Uses structures and pointers to manage satellite data.
- Uses functions to modularize packet processing.
- Uses POSIX threads to process multiple telemetry streams concurrently.
- Uses mutex/semaphore synchronization to prevent simultaneous updates to shared telemetry data.
- Stores processed telemetry in a file for later analysis.
- Uses command-line arguments to specify satellite ID and input/output files.
- Uses IPC or socket programming to simulate communication between the satellite simulator and ground station.

Hackathon Challenge: The system must continue processing telemetry even when multiple packets arrive at almost the same time without losing or corrupting records.

2. A hospital emergency department receives patient requests from multiple registration counters simultaneously. Each patient record contains patient ID, emergency level, symptoms, assigned doctor, and treatment status.

Develop a Smart Emergency Patient Coordination System in C that:

- Dynamically allocates memory for patients.
- Uses functions for registration, search, update, and discharge.
- Uses pointers to modify patient records.
- Uses threads to simulate multiple hospital departments.